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| skkmigas PERTAMINA EP | **PERTAMINA EP REGIONAL 1 ZONA 4** | POG |

| | |
|---|---|
| **MECHANICAL DATASHEET FOR TEST PUMP KRG (P-003 A/B)** | No. Dok. : TMBP-SP001-ME-DST-D115<br>Rev. : A<br>Halaman : 1 dari 14 |

**NAMA PEMILIK PROYEK** : **PT PERTAMINA EP**

**NAMA PROYEK** : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

**LOKASI** : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

**NO. KONTRAK** : **SA02035383B**

Genting Oil Kasuri Pte.Ltd.
1 APPROVED WITHOUT COMMENT

| REV. | TGL. | DESKRIPSI | DISIAPKAN | DIPERIKSA | DISETUJUI | DIPERIKSA | DISETUJUI |
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| A | 19/12/2025 | Issued For Review | ZLN | AS | RTN | | |
| | | | PT. PROSYMPAC OIL & GAS | | | PT. PERTAMINA EP | |

**TABEL REVISI**

| Rev. | Tanggal | Deskripsi |
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## COMMENT RESPONSE SHEET

| No. | Page | Company's Comment | Contractor's Response | Remarks |
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# 1. GENERAL

## 1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

| ID | Easting (Meter) | Northing (Meter) | Elevation (Meter) | Feature Code |
|---|---|---|---|---|
| PN-1531 | 424962,745 | 967376,210 | 10,414 | |
| PM-8593 | 425000,093 | 907528,777 | 19,046 | |

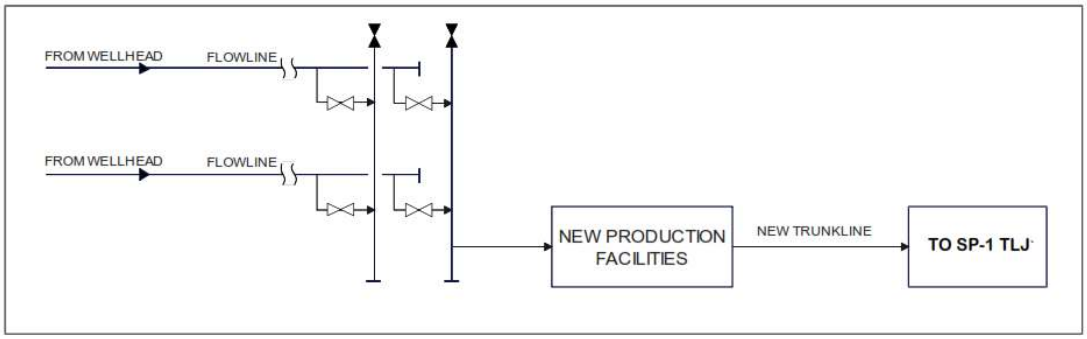
**Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur**

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat

menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

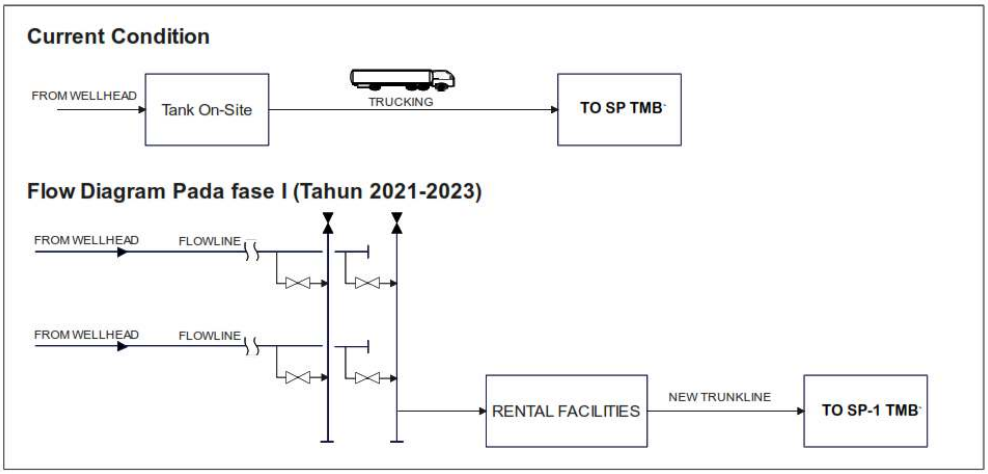
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.




**Gambar 2. Rencana Pola Operasi Fase**

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.




**Gambar 3. Rencana Pola Operasi Fase**

## 1.2 Tujuan

Dokumen ini menjelaskan dipersiapkan sebagai Mechanical Datasheet Test Pump KRG (P-003 A/B) sebagai bagian dari desain yang harus diikuti oieh KONTRAKTOR dan VENDOR untuk proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat- Karangan Timur, Field Prabumulih.

## 1.3 Ruang Lingkup

Mechanical Datasheet ini meliputi ketentuan minimal dalam mengatur desain. material,pabrikasi, pemeriksaan, pengujian, persiapan, pengecatan dan persiapan untuk pengiriman Test Pump KRG (P-003 A/B) untuk dipasang di proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat — Karangan Timur, Field Prabumulih.

## 1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

| | |
|---|---|
| Perusahaan | Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir. |
| Kontraktor | Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN. |
| Vendor | Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR. |

| | |
|---|---|
| SKK MIGAS | Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas. |
| Perencana | Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order. |
| Inspektor | Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN |

## 2. KODE, STANDAR, DAN REFERENSI

DOKUMEN PROYEK

1. TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
2. TMBP-02-PR-CAL-303 — Process Calculation Test Pump KRG (P-003 A/B)
3. TMBP-SP001-PR-DST-D107 —Process Datasheet – Test Pump P-002 & P-003 A/B
4. TMBP-01-PR-PID-104 —Piping and Instrument diagram for Test Tank KRG (T-004) & Test Pump KRG (P-003 A/B)
5. TMBP-SP001-ME-SPE-D103 — Specification – API Centrifugal Pump

## 3. Daftar Singkatan

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

| | |
|---|---|
| **BOPD** | *Barrel Oil Per Day* |
| **BPD** | *Barrel Per Day* |
| **BWPD** | *Barrel Water Per Day* |
| **EOR** | *Enhanced Oil Recovery* |
| **FEED** | *Front End Engineering Design* |
| **KRG** | *Karangan* |
| **MMSCFD** | *Million Standard Cubic Feet per Day* |
| **NPS** | *Nominal Pipe Size* |
| **PB** | *Production Battery* |

| | |
|---|---|
| **PHR** | *Pertamina Hulu Rokan* |
| **PT** | *Perseroan Terbatas* |
| **SP** | *Stasiun Pengumpul* |
| **SCH** | *Schedule* |
| **TMB** | *Tanjung Miring Barat* |
| **TMT** | *Tanjung Miring Timur* |
| **WK** | *Wilayah Kerja* |

# JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

skkmigas PERTAMINA EP

| Title : | Document No. | TMBP-SP001-ME-DST-D115 |
|---|---|---|
| MECHANICAL DATASHEET FOR TEST PUMP KRG (P-003 AB) | Revision | A |
| | Issued Date | 19-Des-25 |
| | Page | 10 OF 14 |

## GENERAL

| | | | | |
|---|---|---|---|---|
| APPLICABLE TO: | PROPOSAL | APPLICABLE NATIONAL / INTERNATIONAL STANDARD : | API-610 (Note 3) | Rev |
| CLIENT | Pertamina EP Regional 1 Zona 4 | UNIT | | |
| SITE | Fasilitas Produksi Stasiun Pengumpul (SP) | SERVICE | Test Pump KRG | |
| NO. REQ 2x100% (Note 12) | PUMP SIZE * | TYPE | OH2 | No. STAGES 1 (One) |
| MANUFACTURER | * | MODEL | * | SERIAL NO. * |
| EQUIPMENT NUMBER | P-003 A/B | | | |

## LIQUID CHARACTERISTICS

| | Units | | | | | | |
|---|---|---|---|---|---|---|---|
| LIQUID TYPE OR NAME : | HC Condensate | | | | | | |
| | | Rated | Normal | MAX | MIN | | |
| LIQUID TYPE OR NAME : | | Groos Oil and Water | | | | | |
| VAPOR PRESSURE : | (psi a) | 6.66 | | | | | |
| SG @ Pump. Temp (3.1.53) : | | 0.97 | | | | | |
| SPECIFIC HEAT : | (Btu/lb) | | | | | | |
| VISCOSITY @ Pump. Temp. : | (cP) | 9.71 | | | | | |

## OPERATING CONDITIONS (6.1.3)

| | Units | Rated | Normal | MAX | MIN | Alt Condition 1 {Name 3} | Alt Condition 2 {Name 4} |
|---|---|---|---|---|---|---|---|
| NPSHA Datum: | C.L. Impeller | | | | | | |
| PUMPING TEMPERATURE : | (ºF) | 101 | | | | | |
| FLOW (Note 11) : | (BPD) | 2300 | | | | | |
| DISCHARGE PRESSURE (6.3.2) : | (psig) | 39.00 | | | | | |
| SUCTION PRESSURE : | (psig) | 0.03 | | | | | |
| DIFFERENTIAL PRESSURE : | (psi) | 38.97 | | | | | |
| DIFFERENTIAL HEAD : | (ft) | 92.80 | | | | | |
| NPSHA (Note 1) : | (ft) | 19.32 | | | | | |
| HYDRAULIC POWER: | (hp) | 1.57 | | | | | |

| | | | |
|---|---|---|---|
| SERVICE (CONTINUOUS/INTERMITTENT) : | INTERMITTENT | INSTALLATION LOCATION : | OUTDOOR |
| • IF INTERMITTENT, NO. OF STARTS : | | IF INDOOR, TEMPERATURE: MAX: ºC | MIN: ºC |
| PUMPS OPERATE IN: | ALONE | ELECTRIC AREA CLASSIFICATION (6.1.29): | HAZARDOUS |
| CORROSION DUE TO : (6.12.1.9) | N/A | CLASS | ZONE 2 |
| EROSION DUE TO : (6.12.1.9) | | GROUP | IIA |
| H2S CONCENTRATION (ppm) : (6.12.1.13) | N/A | TEMP CLASS | T3 |
| CHLORIDE CONCENTRATION | | PUMP VALVE START POSITION : | Open Valve (Fully Loaded) |
| PARTICULATE SIZE (DIA IN MICRONS) | Note 1 | | |
| PARTICULATE CONCENTRATION (ppm) | N/A | | |
| CORROSION ALLOWANCE (6.3.10) : | | | |
| DESIGN NOTES: | Tropicalization Required | | |

## PERFORMANCE

| | | | | |
|---|---|---|---|---|
| PROPOSAL CURVE NO. | * | RPM | * | |
| IMPELLER DIA.: RATED | * | MAX. * | MIN. * | (in) |
| RATED POWER : | * HP | EFFICIENCY | * | % |
| RATED CURVE BEP FLOW (at rated impeller diameter) (6.1.16) : | | | * | (bpd) |
| MIN FLOW : THERMAL | * (bpd) | STABLE | * | (bpd) |
| PREFERRED OPERATING REGION (6.1.16) | * | to | * | (bpd) |
| ALLOWABLE OPERATING REGION (6.9.4.1) | * | to | * | (bpd) |
| MAX HEAD @ RATED IMPELLER | | | * | (ft) |
| MAX POWER @ RATED IMPELLER | | | * | (HP) |
| NPSH3 AT RATED FLOW (6.1.9) : | | | * | (ft) |
| CENTERLINE OF PUMP TO NPSHA DATUM | | | * | (ft) |
| NPSH MARGIN AT RATED FLOW (6.1.10) : | | | * | (ft) |
| SPECIFIC SPEED (6.1.17) | | m3/h, rpm, m | * | (m3/h, rpm, m) |
| SUCTION SPECIFIC SPEED LIMIT (6.1.11) | | m3/h, rpm, m | * | |
| SUCTION SPECIFIC SPEED (6.1.11) | | m3/h, rpm, m | * | (m3/h, rpm, m) |
| MAX. ALLOWABLE SOUND PRESSURE LEVEL (6.1.19) | | | 85 @1 m | (dBA) |
| ESTIMATED MAX. SOUND PRESSURE LEVEL | | | * | (dBA) |
| MAX. ALLOWABLE SOUND POWER LEVEL (6.1.19) | | | | (dB) |
| ESTIMATED MAX SOUND POWER LEVEL | | | * | (dB) |

## DRIVER (7.1) (Note 4)

| | |
|---|---|
| DRIVER TYPE | Electric Motor |
| GEAR | NO |
| VARIABLE SPEED REQUIRED | NO |
| SOURCE OF VARIABLE SPEED | N/A |
| MANUFACTURER | * |
| NAMEPLATE POWER : HP | (Note 12) |
| NOMINAL RPM | * |
| RATED LOAD RPM | * |
| FRAME | * |
| ORIENTATION | HORIZONTAL |
| LUBRICATION | * |
| BEARING TYPE | Anti Friction |
| RADIAL | * |
| THRUST | * |
| STARTING METHOD | DOL |
| DRIVER DATASHEET NUMBER | |
| DRIVE DESIGN STANDARD | IEC |

## CONSTRUCTION

API PUMP TYPE (Table 1): OH2

NOZZLE CONNECTIONS (6.4.2):

| | Size | Facing | Rating | Position |
|---|---|---|---|---|
| SUCTION | 3" (*) | RF | 300 | END |
| DISCHARGE | 2" (*) | RF | 300 | TOP |

PRESSURE CASING AUX. CONNECTIONS: (6.4.3)

| | No. | Size | Type | Facing | Rating | Posn. |
|---|---|---|---|---|---|---|
| BALANCE/LEAK OFF | | | | | | |
| DRAIN (Note 6) | | (*) | | RF | 300 | |
| VENT (Note 6) | | (*) | | RF | 300 | |
| PRESSURE GAGE | | | | | | |
| TEMP GAGE | | | | | | |
| WARM-UP LINE | | | | | | |

| | |
|---|---|
| DRAIN VALVE SUPPLIED BY (7.6.2.5) | SUPPLIER |
| DRAINS MANIFOLDED (7.6.2.5) | NO |
| VENT VALVE SUPPLIED BY | SUPPLIER |
| VENTS MANIFOLDED | NO |
| THREADED CONS FOR PIPELINE SERVICE & < 55°C (6.4.3.2) | |
| SPECIAL FITTINGS FOR TRANSITIONING (6.4.3.3) | |
| CYLINDRICAL THREADS REQUIRED (6.4.3.2) | |
| MACHINED AND STUDDED AUX CONNECTIONS (6.4.3.6) | |
| ROUTE DRAIN TO SKID EDGE | YES |

### MATERIAL (6.12.1.1) (Note 8)

| | | |
|---|---|---|
| ANNEX H CLASS | D-1 | |
| MINIMUM DESIGN METAL TEMPERATURE (6.12.4.1) | 0 | °C |
| MAXIMUM ALLOWABLE TEMPERATURE (3.1.19) | | °C |
| REDUCED-HARDNESS MATERIALS REQ'D (6.12.1.14) | | |
| APPLICABLE HARDNESS STANDARD (6.12.1.14) | | |
| BARREL : | | |
| CASE : | Duplex | |
| DIFFUSERS | | |
| IMPELLER : | Duplex | |
| IMPELLER WEAR RING : | Hard Faced Duplex | |
| CASE WEAR RING : | Hard Faced Duplex | |
| SHAFT: | Duplex | |
| BOWL (IF VS-TYPE) | | |

### BEARINGS AND BEARING HOUSINGS (6.10.1)

| | | | |
|---|---|---|---|
| BEARING (TYPE / NUMBER): | | | |
| RADIAL | * | / | * |
| THRUST | * | / | * |
| REVIEW AND APPROVE THRUST BEARING SIZE : (9.2.5.2.6) | | | |
| LUBRICATION (6.11.3) (9.3.12.4) : (Note 11, 14) | | | * |
| PRESSURE LUBE SYSTEM TO API-614, CHAPTER : | | | * |
| API 614 DATASHEETS ATTACHED (9.2.6.4) | | | * |
| PRESSURIZED LUBE OIL SYSTEM MOUNTED ON BASEPLATE: | | | * |
| LOCATION OF PRESSURIZED LUBE OIL SYSTEM: | | | * |
| INTERCONNECTING PIPING PROVIDED BY | | | * |
| OIL VISC. ISO GRADE | | | * |
| VENT-TO-HOUSING CONSTANT LEVEL OILER (6.10.2.4) : | | | * |
| OIL MIST PROVISIONS (6.11.3) | | | N/A |
| GREASE LUBRICATION (6.11.4) | | | N/A |

| | | | | |
|---|---|---|---|---|
| CASING MOUNTING (6.3.14, 9.3.8.3): | CENTERLINE | | | |
| CASING TYPE: | SINGLE VOLUTE | | | |
| OH3 BACK-PULLOUT LIFTING DEVICE REQD. (9.1.2.6) | N/A | | | |
| CASING PRESSURE RATING: | | | | |
| MAWP (6.3.6) : | * | psig | @ | * (ºF) |
| HYDROTEST (8.3.2.2) : | * | psig | @ | AMB (ºF) |
| WETTING AGENT REQUIRED FOR HYDROTEST (8.3.2.7) : | | | | * |
| HYDROTEST OH PUMP AS ASSEMBLY (8.3.2.14) : | | | | * |
| SUCT'N PRESS. REGIONS DESIGNED FOR MAWP (6.3.8) : | | | | YES |

| | |
|---|---|
| ROTOR: | |
| SHAFT FLEXIBILITY INDEX (SFI) (9.1.1.3) | * |
| FIRST CRITICAL SPEED WET (MULTISTAGE PUMPS ONLY) | N / A |
| COMPONENT BALANCE TO ISO 1940-1, G1 (6.9.3.4) : | YES |
| SHRINK-FIT-LIMITED MOVEMENT IMPELLERS (9.2.2.3) | * |
| ROTATION (VIEWED FROM COUPLING END) : | * |
| IMPELLERS INDIVIDUALLY SECURED (6.6.3) : | YES |

| | | |
|---|---|---|
| COUPLING: (7.2) | | |
| MANUFACTURER | * | |
| MODEL | Flexible Type (*) | |
| RATING (kW / 100 RPM ) | * | |
| SPACER LENGTH (7.2.2.d) | * | mm |
| SERVICE FACTOR (7.2.3) | * | |
| RIGID-TYPE | NO | |
| COUPLING WITH HYDRAULIC FIT (7.2.9) | * | |
| COUPLING WITH PROPRIETARY CLAMPING DEVICE (7.2.10) | * | |
| COUPLING IN COMPLIANCE WITH (7.2.4)(7.2.2.f) | API 610 | |
| GUARDS (7.3) | | |
| COUPLING AND SHAFT GUARD STANDARD (7.3) | * | |
| IGNITION HAZARD ASSESSMENT PER EN 13463-1 (7.3.2.2; 7.3.3.4) | * | |
| COUPLING GUARD MATERIAL (7.3.2.1; 7.3.3.3) : | Non-sparking | |
| SHAFT GUARD MATERIAL (7.3.2.1; 7.3.3.3) : | * | |
| SPARK RESISTANT MATERIAL REQUIRED (7.3.2.1) : | YES | |

### BASEPLATE

| | |
|---|---|
| API BASEPLATE NUMBER (ANNEX D) : | * |
| IF NON-STD BASEPLATE DIMENSIONS (LxW) ( | |
| BASEPLATE CONSTRUCTION (7.4) | * |
| BASEPLATE DRAINAGE (7.4.1) | * |
| MOUNTING : | GROUTED |
| NON-GROUT CONSTRUCTION (7.4.1.e): | NOT REQUIRED |
| SUPPLIED WITH : ● GROUT AND VENT HOLES | YES |
| ● DRAIN CONNECTION | YES |
| DEMONSTRATE BASEPLATE PAD FLATNESS (7.4.9) | * |
| PROVIDE SPACER PLATE UNDER ALL EQUIPMENT FEET (7.4.10) | |
| BOLT OH 3/4/5 PUMP TO PAD / FOUNDATION : | N / A |
| PROVIDE SOLEPLATE FOR OH 3/4/5 PUMPS (9.1.2) : | N / A |

REMARKS:

DISCHARGE CIRCUIT PIPING DESIGN:

DESIGN PRESSURE _____ psig

DESIGN TEMPERATURE _____ ºF

Rev

## INSTRUMENTATION

| | |
|---|---|
| INSTRUMENTATION PER API-670 (7.5.2) | N / A |
| ACCELEROMETERS (7.5.2.1) | |
| NUMBER OF ACCELEROMETERS | N/A |
| MOUNTING LOCATION: | |
| PROVISION FOR MTG ONLY (6.10.2.13) | |
| FLAT SURFACE REQUIRED (6.10.2.14) | YES |
| VIBRATION PROBES (7.5.2.2) | |
| VIBRATION PROBES REQUIRED (7.5.2.2) | N / A |
| NUMBER PER RADIAL BEARING | |
| NUMBER PER AXIAL BEARING | |
| THREADED PROVISION FOR MTG ONLY (6.10.2.13; 6.6.12) | |
| FLAT SURFACE PROVISION ONLY (6.10.2.14) | YES |
| MONITORS AND CABLES SUPPLIED BY (7.5.2.4) | |
| TEMPERATURE DETECTORS (7.5.2.3) | |
| TEMP. PROBES REQUIRED (7.5.2.3) | N / A |
| PROVISIONS FOR MOUNTING ONLY (6.10.2.2) | |
| RADIAL BEARING TEMP. | |
| NUMBER PER RADIAL BEARING | |
| THRUST BEARING TEMP. | |
| NUMBER PER THRUST BEARING ACTIVE SIDE | |
| NUMBER PER THRUST BEARING INACTIVE SIDE | |
| TEMP. GAUGES (WITH THERMOWELLS) (9.1.3.5) | |
| TEMP. GAUGE LOCATION | |
| SUPPLY UPPER/LOWER CASING RTD'S FOR WARMUP | |
| PRESSURE GAUGE TYPE | |
| PRESSURE GAUGE LOCATION | |

**REMARKS**

## PIPING & APPURTENANCES

| | |
|---|---|
| TAG ALL ORIFICES (7.6.2.4) | * |
| SOCKET WELD UNION ON 1st SEAL GLAND NIPPLE (7.6.2.8) | * |
| MANIFOLD AUX PIPING SYSTEMS AT SKID EDGE (7.6.1.6) | * |

## SEAL SUPPORT SYSTEM MOUNTING

| | |
|---|---|
| MOUNTED ON PUMP BASEPLATE (7.6.1.4) | N / A |
| LOCATION ON OR OFF BASEPLATE (7.4.6) : | ON BASEPLATE |
| INTERCONNECTING PIPING BY | Supplier |

## MECHANICAL SEAL (6.8) (Note 7)

| | |
|---|---|
| SEAL TYPE | API 682 MECHANICAL SEAL |
| API 682 DATASHEET ATTACHED : | YES |
| ADDITIONAL CENTRAL FLUSH PORT (6.8.9) | |
| HEATING JACKET REQ'D. (6.8.11) | |
| SEAL MANUFACTURER | * |
| SIZE / TYPE | * / * |
| MANUFACTURER CODE | * |
| ARRANGEMENT | DUAL |
| **SEAL CHAMBER DATA:** | |
| TEMPERATURE | * °C |
| PRESSURE | * kg/cm²g |
| FLOW | * m³/h |
| **SEAL FLUSH PIPING PLAN:** | 31 + 52(NOTE 7) |
| * in | * |
| SEAL FLUSH FLUID TYPE | * |
| PIPING ASSEMBLY | BY SUPPLIER |
| **AUXILIARY PIPING PLAN:** | * |
| * in | * |
| AUXILIARY FLUSH FLUID TYPE | * |
| PIPING ASSEMBLY | BY SUPPLIER |

## HEATING AND COOLING (6.1.23-6.1.27)

| | | |
|---|---|---|
| COOLING REQ'D | | |
| COOLING WATER PIPING PLAN (7.6.3.1) | | |
| COOLING WATER PIPING | | |
| FITTINGS | | |
| COOLING WATER PIPING MATERIALS | | |
| COOLING WATER REQUIREMENTS: | SUPPLY PRESSURE | kg/cm²g |
| | BEARING HOUSING | m³/h |
| | HEAT EXCHANGER | m³/h |
| | TOTAL COOLING WATER | m³/h |
| HEATING MEDIUM | N/A | |
| HEATING PIPING | N/A | |

## PRESSURE VESSEL DESIGN CODE REFERENCES

THESE REFERENCES SHALL BE PROVIDED BY THE MANUFACTURER

CASTING FACTORS USED IN DESIGN (PER TABLE 4)

SOURCE OF MATERIAL PROPERTIES (6.3.5)

## WELDING AND REPAIRS (6.12.3.1)

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER. (DEFAULT TO TABLE 11 IF NO PURCHASER PREFERENCE IS STATED)

| | |
|---|---|
| ALTERNATIVE WELDING CODES AND STANDARDS | |
| WELDING REQUIREMENT (APPLICABLE CODE OR STANDARD) | DEFAULT PER TABLE 11 |
| ALTERNATIVE WELDER/OPERATOR QUALIFICATION STANDARD | |
| ALTERNATIVE WELDING PROCEDURE QUALIFICATION STANDARD | |
| NON-PRESSURE RETAINING STRUCTURAL WELDING STANDARD (BASEPLATES OR SUPPORTS) | |
| STANDARD FOR MAGNETIC PARTICLE OR LIQUID PENETRANT EXAMINATION (PLATE EDGES) | |
| STANDARD FOR POSTWELD HEAT TREATMENT | |
| STANDARD FOR POSTWELD HEAT TREATMENT OF CASING FABRICATION WELDS | |

## MATERIAL INSPECTION

THESE REFERENCES SHALL BE PROVIDED BY THE PURCHASER

DEFAULT TO TABLE 14 **YES** TABLE 14 INSPECTION CLASS: **II**

ALTERNATIVE MATERIAL INSPECTIONS AND ACCEPTANCE CRITERIA (SEE TABLE 15, 8.2.2.5)

| TYPE OF INSPECTION | METHOD | FOR FABRICATIONS | FOR CASTINGS |
|---|---|---|---|
| RADIOGRAPHY | | | |
| ULTRASONIC INSPECTION | | | |
| MAGNETIC PARTICLE INSPECTION | | | |
| LIQUID PENETRANT INSPECTION | | | |
| VISUAL INSPECTION (all surfaces) | | | |

## SURFACE PREPARATION, PAINT & SPARES

| | |
|---|---|
| MANUFACTURER'S STANDARD (8.4.3.4) | |
| OTHER (SEE BELOW) | YES |
| SPECIFICATION NO. | Note 5 |
| **PUMP:** | |
| PUMP SURFACE PREPARATION | as per project specification |
| PRIMER | as per project specification |
| FINISH COAT | as per project specification |
| **BASEPLATE:** | |
| SURFACE PREPARATION | as per project specification |
| PRIMER: | as per project specification |
| FINISH COAT | as per project specification |
| UNDERSIDE | as per project specification |
| **SHIPMENT : (8.4)** | |
| EXPORT BOXING REQUIRED | |
| PREPARE FOR OUTDOOR STORAGE (NUMBER OF MONTHS) : | |
| **SPARE ROTOR ASSEMBLY PACKAGED FOR:** | |
| ROTOR STORAGE ORIENTATION (9.2.8.2) | |
| SHIPPING & STORAGE CONTAINER SUITABLE FOR VERTICAL STORAGE (9.2.8.3) : | |
| N2 PURGE REQUIRED (9.2.8.4) : | |
| **SPARE PARTS (10.3.4.2)** | |
| START-UP | YES |
| NORMAL MAINTENANCE | |

**MASSES** kg

| ITEM No | PUMP | DRIVER | GEAR | BASE | TOTAL |
|---|---|---|---|---|---|
| **P-003 A/B** | * | * | * | * | * |
| | | | | | |
| | | | | | |

## OTHER PURCHASER REQUIREMENTS

| | |
|---|---|
| COORDINATION MEETING REQUIRED | YES |
| MAXIMUM DISCHARGE PRESSURE TO INCLUDE (6.3.2) | |
| MAX RELATIVE DENSITY | |
| OPERATION TO TRIP SPEED | |
| MAX DIA. IMPELLERS AND/OR NO OF STAGES | |
| CONNECTION DESIGN APPROVAL (9.2.1.4) | |
| TORSIONAL ANALYSIS / REPORT (6.9.2) | |
| PROGRESS REPORTS (L.3.3) | YES |
| OUTLINE OF PROCEDURES USED FOR OPTIONAL TESTS (L.2.5) | |
| ADDITIONNAL DATA REQUIRING 20 YEARS RETENTION (8.2.1.1) | |
| LATERAL ANALYSIS REQUIRED (9.2.4.1.2) | |
| ROTOR DYNAMIC BALANCE TO 4W/N (6.9.3.5) | |
| VFD STEADY STATE FORCED RESPONSE ANALYSIS (6.9.2.3) | |
| TRANSIENT FORCED RESPONSE (6.9.2.4) | |
| BEARING LIFE CALCULATIONS REQUIRED (6.10.1.11) | YES |
| CASING RETIREMENT THICKNESS DRAWING [L.3.6.1 cc) 7) xiii)] | |
| CONNECTION BOLTING (7.6.1.7) | |
| VENDOR TO KEEP REPAIR AND HT RCDS (8.2.1.1.c) | YES |
| VENDOR SUBMIT TEST PROCEDURES (8.3.1.1) | YES |
| SUBMIT INSPECTION CHECK LIST (8.1.5) | YES |
| ACOUSTIC ANALYSIS OF CROSSOVER PASSAGE (BB3, BB5)(9.2.1.5) | |
| API-691 DOCUMENTATION REQUIRED (6.1.3.1) | |

## TEST

| | | Rev |
|---|---|---|
| SHOP INSPECTION (8.1.1.1) | YES | |
| PERF. CURVE & DATA APPROVAL REQ'D PRIOR TO SHIPMENT (8.3.3.4.6) : | | |
| TEST WITH SUBSTITUTE SEAL (8.3.3.3.1) | | |
| MATERIAL CERTIFICATION REQUIRED (6.12.1.8): **(Note 2** CASING | YES | |
| IMPELLER | YES | |
| SHAFT | YES | |
| OTHER | | |
| CASTING REPAIR WELD PROCEDURE APPROVAL REQD (6.12.2.5): | YES | |
| INSPECTION REQUIRED FOR CONNECTION WELDS (6.12.3.4) | | |
| MAG PARTICLE | | |
| RADIOGRAPHY | | |
| LIQUID PENETRANT | YES | |
| ULTRASONIC | | |
| HARDNESS TEST REQUIRED (6.12.1.14; 8.2.2.7) | | |
| ADDITIONAL SUBSURFACE EXAMINATION (6.12.1.6) (8.2.1.3) | | |
| FOR: | Casting | |
| METHOD: | | |
| PMI TESTING REQUIRED (8.2.2.8) | YES | |
| COMPONENTS TO BE TESTED : | For Alloy Material | |
| RESIDUAL UNBALANCE TEST (J.4.1.2) | | |
| NOTIFICATION OF SUCCESSFUL PERFORMANCE TEST (8.1.1.3) (8.3.3.4.6) | | |
| BASEPLATE (NOZZLE LOAD) TEST (7.4.24) | | |
| HYDROSTATIC TEST (8.3.2) | OBSERVE | |
| HYDROSTATIC TEST OF BOWLS & COLUMN (9.3.13.1) | | |
| PERFORMANCE TEST (8.3.3) | WIT | |
| RETEST ON SEAL LEAKAGE (8.3.3.3.2) | OBSERVE | |
| TEST DATA POINTS TO (8.3.3.4) | | |
| ALTERNATE TEST TOLERANCES PER (8.3.3.5) | | |
| NPSH (8.3.4.3) (if margin < 1m (3ft) | WIT | |
| NPSH BASED ON 1ST STG ONLY ALLOWED (8.3.4.3.2) | N/A | |
| TEST NPSHA LIMITED TO 110% SITE NPSHA (8.3.3.7) | | |
| RETEST REQUIRED AFTER FINAL HEAD ADJUSTMENT (8.3.3.8.2) | | |
| COMPLETE UNIT TEST (8.3.4.4.1) **(Note 9)** | WIT | |
| SOUND LEVEL TEST (8.3.4.5) | WIT | |
| CLEANLINESS PRIOR TO FINAL ASSEMBLY (8.2.2.6) | NON-WIT | |
| LOCATION OF CLEANLINESS INSPECTION: | | |
| CHECK FOR CO-PLANAR MOUNTING PAD FLATNESS (7.4.8) | | |
| MECHANICAL RUN TEST UNTIL OIL TEMP STABLE (8.3.4.2.1) | | |
| 4 HR. MECH RUN AFTER OIL TEMP STABLE (8.3.4.2.2) | WIT | |
| BRG HSG RESONANCE TEST (8.3.4.7) | | |
| STRUCTURAL RESONANCE TEST (9.3.9.2) | | |
| REMOVE / INSPECT HYDRODYNAMIC BEARINGS AFTER TEST (9.2.7.4) | | |
| AUXILIARY EQUIPMENT TEST (8.3.4.6) | NON-WIT | |
| EQUIPMENT TO BE INCLUDED IN AUXILLIARY TESTS : | | |
| LOCATION OF AUXILIARY EQUIPMENT TEST : | | |
| IMPACT TEST (6.12.4.3) PER EN 13445 | | |
| PER ASME SECTION VIII | | |
| CASING DISASSEMBLY AFTER TEST (8.3.3.9) | | |
| SPARE PARTS TEST (8.3.4.8) | | |

Rev.

## NOTES

(*) = VTA (VENDOR to advise and shall fill detail value or data before Purchase Order)

1 Margin NPSH shall be minimum 1 meter (3 ft) otherwise Witness NPSH Test is required.

2 Material certificates shall be provided for 3.1 for wetted parts & pressure parts and 2.2 for others as per ISO 10474 Standard.

3 The Pump shall be designed as per below requirements:
- TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
- TMBP-02-PR-CAL-303 — Process Calculation Test Pump KRG (P-003 A/B)
-TMBP-SP001-PR-DST-D107 —Process Datasheet – Test Pump P-002 & P-003 A/B
- TMBP-01-PR-PID-104 —Piping and Instrument diagram for Test Tank KRG (T-004) & Test Pump KRG (P-003 A/B)
- TMBP-SP001-ME-SPE-D103 — Specification – API Centrifugal Pump
- API 610 Latest Edition — Centrifugal Pumps for Petroleum, Petrochemical and Natural Gas Industries
- API 682 — Pumps – Shaft Sealing Systems for Centrifugal and Rotary Pumps

4 Motor shall be designed as per below requirements:
- Motor shall be as per IEC Standard & shall be sized for End of Curve (EOC) operation
- TMBP-SP001-EL-SPE-D106 — Specification for Induction Motor

5 Finish/top color shall comply as per Document Painting & Coating TMBP-SP001-PI-SPE-D109 “Piping Painting & Lettering Specification”.

6 Drain & Vent connection shall be provided with isolation valve and blind flange.

7 Mechanical seal shall be designed as per below requirement:
- Mechanical seal shall be compatible with process liquid
- VENDOR shall check, design and guarantee that material of mechanical seal is suitable as per process liquid
- VENDOR shall provide mechanical seal selection from seal manufacturer before Purchase Order (P.O)

8 VENDOR shall check and guarantee that material of the pump is suitable as per process liquid before Purchase Order (P.O).

9 Complete Unit Test of the pump shall include with job motor & job mechanical seal.

10 Pump should be completed with oil level glass.

11 Pump actual capacity shall be ±10% from design capacity.

12 Pump operation shall be 2×100%, 1 (one) running and 1 (one) standby.

13 VENDOR shall recalculate actual power rating before Purchase Order (PO).

14 Temperature of oil at bearing chamber shall not exceed 140 °F and max 180 °F will shut down the pump.

15 VENDOR shall include below requirement:
- Factory Acceptance Test (FAT) execution
- Motor solo run, site realignment, SAT including commissioning supervision
- Certificate material 3.1 and 3.2 for critical part shaft and impeller
- First Filling: Lube oil/Grease
- Mating Flange, Counter Flange (if required)
- Insulation for bolt connection at flange suction and discharge